TravelSense: A Personalized Travel Recommendation and Adaptive Itinerary Planning System Using Heterogeneous Graph Neural Networks

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ABSTRACT

Nowadays, creating a good travel plan has become somewhat difficult for individuals; there are so many recommendations tailored to various interests that it becomes hard to decide where to go. A traveler must consider several factors, such as time, budget, accommodation, and the destination itself. To make recommendations more accurate, various methods are employed—including content-based filtering, collaborative filtering, and personalized approaches. AI and machine learning tools—specifically Graph Neural Networks—are used to analyze a user's past preferences and areas of interest. Traditional collaborative filtering often struggles with issues like data sparsity, the "cold start" problem, and handling less popular items, which can lead to suboptimal predictions. The proposed system considers factors such as the type of trip, group size, duration of the stay, and the user's personal interests to suggest places to stay, dine, and visit. Consequently, there is a need for an effective travel recommendation system—powered by advanced technology—that simplifies the decision-making process and provides practical solutions to real-world travel challenges.

Index Terms—Graph Neural Network, content-based filtering, personalized travel recommendation, Machine learning model

1. INTRODUCTION

Nowadays, there are so many places to visit and activities to engage in that a traveler can easily become confused about where to go to make the best use of their time. No travel recommendation app can perfectly match a traveler's specific interests every single time, as not every activity is feasible at every location. While it is sometimes possible to find a perfect match, it requires aggregating data from various sources and tailoring it to the user's individual preferences. Everyone loves going on vacation, but planning one is often time-consuming. We have to examine the ratings, reviews, and features of hotels, attractions, and restaurants, while also ensuring they fit within our budget. Social media frequently showcases new destinations, which can inspire a traveler to set off on a trip. While it is easy to discover potential travel spots—whether through friends, social media, or other channels—the challenge lies in selecting the *best* place. Travelers often struggle to gauge whether a particular destination will suit them or what kind of experience they might have there. Travel recommendation systems are designed to address these issues; while they may not eliminate confusion entirely, they significantly reduce it by suggesting destinations that align with a traveler's personal interests. A travel recommendation system is a tool that tourists and travelers use to meet their specific needs. It simplifies the process of choosing a destination, discovering nearby points of interest (POIs) and restaurants, determining the shortest routes, and finding the best places to stay. There are two primary types of travel recommendation systems: generic and personalized. Generic recommendation systems give you information about places to visit, like tourist attractions nearby Kashmir or Indonesia. On the other hand, personalized recommendation systems need information about the user, like their gender, the type of vacation they want, the number of people, and the length of the trip. Many models are correct, but they only focus on a few well-known landmarks and don't use geotagged data to make personalized suggestions based on what the user likes.

A. Scope

Building a travel recommendation system is a big job that involves many parts of the travel industry and user preferences:

1) User item recommendation: Suggesting travel destinations, places to stay, things to do, restaurants, and other travel-related services based on what other users with similar interests and behaviors like.
2) User to user interest: If you are traveling as a group, you must ensure that everyone's preferences align.
3) Review and Rating: Furthermore, it is important to consider ratings, identify the best places to visit, and look at user reviews for various locations.
4) User profile making: Furthermore, it is necessary to understand each user's interests—keeping track of the places they are drawn to—so that other similar locations can be recommended to them accordingly.

2. LITERATURE REVIEW

Previous technologies have also utilized Graph Neural Networks, yet they failed to deliver the expected results. The ongoing challenge has been to address travel-related issues in a way that not only recommends destinations tailored to a user's personal preferences but also ensures efficiency in terms of both time and cost. However, progress in this area has remained limited; people often remain unsure about where they should actually go. Even when they have a destination in mind, they struggle to understand why it is the right choice—specifically, what activities or attractions exist there that would truly enhance their travel experience? People often travel from one place to another not just for the sake of traveling, but because they have specific needs—such as visiting a hospital, attending work, or going to an interview. The recommendation system we are discussing today is designed to address these types of scenarios. Given the numerous constraints involved, the problem must be broken down into smaller, manageable parts. This allows each component to be addressed individually, making it easier to pinpoint exactly where challenges arise. For instance, determining the optimal route is a key concern—ensuring a smooth journey without complications. While an app might recommend a destination that a user selects, our role extends beyond mere selection; we must also ensure that the entire journey proceeds as planned. Only when these aspects are accounted for can a model be truly effective for travel-related purposes.

A. Traditional Recommendation Techniques

Recommendation systems play a significant role in assisting users. Nowadays, the volume of travel-related data has become so vast that people often find themselves confused about where they should actually go. Recommendation systems help by suggesting destinations tailored to a user's interests, while also taking into account their budget, time constraints, and the various activities they might wish to pursue. Traditional recommendation techniques primarily include content-based filtering, collaborative filtering, and hybrid recommendation methods. In content-based filtering, a user specifies their interests—such as the types of places they enjoy visiting—and the system then recommends locations based on those preferences. For example, if a user prefers mountain destinations and adventure activities, the system can suggest other places with similar characteristics. Essentially, the system operates using data provided by the user to generate location recommendations tailored to their interests.

B. Recommendation Systems in Tourism

A tourism recommendation system gathers information from multiple sources because travel planning involves a series of interconnected decisions. Travelers typically consider various aspects, such as destinations, accommodations, transportation, food, activities, and the local culture and events. Furthermore, their choices are not limited to these factors alone; they must also account for their budget, the timing of their trip, the size of their group, and a wide range of potential activities. This presents challenges, such as ensuring personalization and managing the availability of tourism-related data. This research proposes solutions to address these various issues.

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C. Graph-Based Recommendation Systems

Graph based recommendation system kaa kaam yahi hai ki wo apne nodes ke through information ko show karta hai yaa esse bhi keh sakte hai ki sari user ki past aur present informations ko store karke rakhta hai. Ab usme jo edges hoti hai wo ye represent karti hai ki interactions or relationship between them. Ab iske help se recommendation system information ko read karke user ko interactive places ke ware me show karta hai aur ye bhi batata hai ki ye wo agar ye sab places show kar raha hai to kyu kar raha hai. Graph based recommendation system ek achha potential rakhta hai recommendation ke kaam me. Ab isme ye achha isliye mana gaya hai kyuki go ye system hai wo data ko khud hi read karta hai aur khud hi sab kuch handle karta hai aur uske baad plan ko recommand karta hai.

For example, a destination may be associated with hotels, restaurants, adventure activities, tourist attractions, and local events. Similarly, a user's interest in adventure tourism may be connected to destinations that provide trekking, rafting, or other outdoor activities.

D. Graph Neural Networks for Personalized Recommendation

A graph neural network is a machine learning model that learns representations from graph-structured data. It typically updates the representation of a given node by incorporating information from its neighboring nodes. This process enables the model to learn patterns and understand the relationships between entities. For instance, a graph neural network can learn the relationships between users and destinations while considering other connected entities, such as hotels, activities, and food places. For example, if a user frequently interacts with destinations associated with trekking and adventure sports, the model may learn patterns that help rank similar destinations. The quality of such recommendations depends on the graph structure, available data, model design, and evaluation process.

3. METHODOLOGY

The main objective of TravelSense is to make travel planning more personalized and convenient for users. The proposed system considers user preferences, budget, travel duration, and interests while recommending suitable destinations and related travel options. Instead of treating each travel option independently, the system aims to use the relationships between destinations, hotels, activities, food places, events, and other travel-related entities.

The proposed methodology consists of data collection and preprocessing, heterogeneous travel graph construction, graph neural network-based recommendation, and itinerary generation. Each stage contributes to the overall process of generating a travel plan according to the user's requirements.

The proposed workflow starts with collecting user preferences and travel information. The collected information is processed and used to construct a heterogeneous travel graph. The recommendation component then identifies suitable travel options, which are used to prepare a personalized itinerary. User feedback may also be considered to improve future recommendations when the corresponding functionality is available.

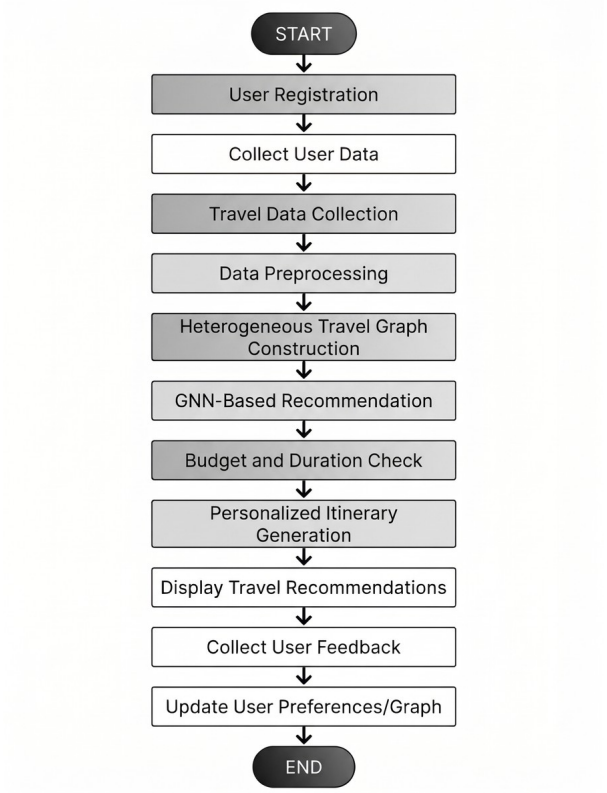


Fig. 1. Overall methodology of the proposed TravelSense system.

A. System Overview

TravelSense follows a modular design in which the user interface, backend services, database, and recommendation component work together to support travel planning. The user provides information such as preferred destinations, budget, trip duration, group size, and desired activities through the interface.

The backend processes these inputs and retrieves the relevant travel information from the database or other available data sources. The recommendation component uses the available information to identify suitable destinations and related options, such as hotels, activities, food places, and events. These options are then considered during itinerary generation.

This modular approach separates user interaction, data management, recommendation, and itinerary planning. It also allows individual components to be developed and improved without requiring the entire system to be redesigned.

The proposed architecture of TravelSense is described as follows. The frontend collects user inputs, while the backend coordinates data retrieval and recommendation requests. The database stores the available user and travel information, and the recommendation component processes the relevant data to identify suitable travel options.

B. Data Collection and Preprocessing

TravelSense requires travel-related information and user preference data to generate relevant recommendations. Travel information may include destination details, tourist attractions, hotels, activities, food locations, events, and transportation options. Depending on availability, these details can be obtained from public datasets, reliable travel information sources, or external APIs.

User preferences may include budget range, travel duration, group size, preferred activities, and destination interests. Where available, user interactions such as viewing, saving, rating, or selecting a destination can provide additional information about travel interests.

Before the data is used, it needs to be organized into a consistent format. This stage may involve removing duplicate records, handling missing values, and standardizing the categories of travel entities. The resulting data provides the foundation for constructing the travel graph and generating recommendations.

The quality of the recommendations depends partly on the accuracy, completeness, and relevance of the available data.

C. Heterogeneous Travel Graph Construction

TravelSense uses a heterogeneous graph to represent the relationships between users and different travel-related entities. A heterogeneous graph is suitable for this purpose because travel planning involves several types of entities that are connected in different ways.

The graph may contain nodes representing users, destinations, hotels, activities, food places, events, and transportation options. Edges represent the relationships between these nodes. For example, a user may prefer a particular destination, a destination may contain several attractions, and an attraction may be associated with one or more activity categories.

User interactions can also be represented as graph relationships when the relevant data is available. For instance, saving or rating a destination may provide information about a user's interests. Different relationship types help distinguish between user preferences and connections among travel entities.

This representation allows the recommendation component to consider information from connected entities instead of relying only on the individual features of a destination. The resulting graph serves as the input structure for the proposed graph-based recommendation approach.

The graph connects different travel entities through meaningful relationships. These connections provide contextual information that can help the recommendation model identify relevant travel options.

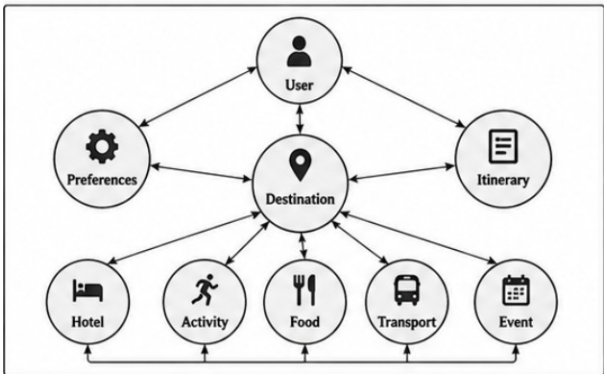


Fig. 2. Conceptual representation of the heterogeneous travel graph used in TravelSense.

D. GNN-Based Recommendation

The proposed recommendation component uses a Graph Neural Network (GNN) to learn representations of users and travel-related entities from the heterogeneous graph. A GNN aggregates information from connected nodes and uses the graph structure to learn relationships between different entities.

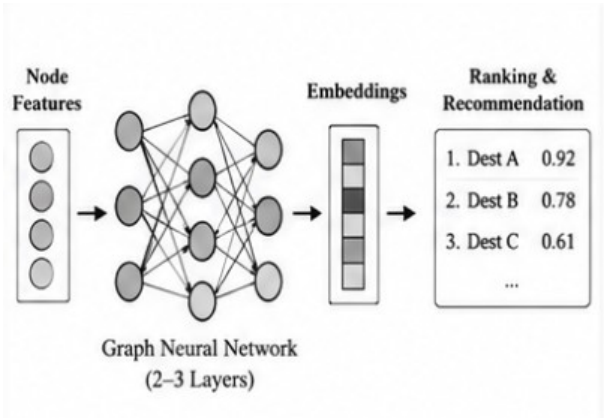


Fig. 3. Proposed GNN-based travel recommendation workflow.

The recommendation process begins by identifying the user's preferences and travel requirements. The model can then use the available graph structure and learned representations to estimate the relevance of candidate destinations and other travel options. These candidates can be ranked to identify options that are more suitable for the user.

For existing users, available interaction history may provide additional information about their interests. For new users with limited interaction history, explicitly provided preferences, such as budget, preferred activities, and travel duration, can serve as the initial basis for recommendations.

The final recommendation list should also consider relevant travel constraints. For example, a destination that matches the user's interests may still be unsuitable if it exceeds the specified budget or does not fit the available travel duration. Therefore, recommendation relevance and practical constraints should be considered together.

The proposed recommendation process begins with the travel graph and ends with ranked recommendations. The effectiveness of this approach depends on the graph representation, available data, model design, and training process.

E. Adaptive Itinerary Generation

After suitable travel options have been identified, TravelSense aims to organize them into a personalized itinerary. The itinerary-generation process considers the user's selected destination, budget, trip duration, group size, and preferred activities.

The system can select suitable attractions and activities and arrange them into a structured travel plan. When location and timing information is available, geographical proximity and estimated travel time can also be considered while organizing visits. Hotels, food places, events, and transportation options may be included according to the user's requirements and the available information.

The generated itinerary is intended to provide a more organized travel experience than a simple list of recommended places. If the user changes their preferences or travel requirements, the itinerary can be regenerated using the updated information.

The feasibility of the itinerary depends on the quality of the available location, timing, cost, and availability data. These factors should be considered when evaluating the generated travel plan.

F. User Feedback and Recommendation Adaptation

User feedback can provide additional information about the usefulness of travel recommendations. TravelSense is designed to consider interactions such as saving a destination, rating a place, or selecting an option for an itinerary, provided that these interactions are collected by the system.

Such feedback can help identify whether a user is interested in particular destinations or activities. When the feedback mechanism is implemented, this information can be used to update user preference data and support future recommendations.

For example, a user who repeatedly selects cultural attractions may receive more recommendations related to cultural activities. However, the actual effect of feedback on recommendation quality needs to be evaluated using appropriate data and testing. This mechanism is intended to make the system more responsive to changing travel interests.

G. Implementation Considerations

The proposed framework can be implemented using a web-based frontend, backend services, a database, and a graph-based recommendation component. FastAPI can be used to develop backend APIs, NetworkX can support graph construction and exploration, and PyTorch Geometric can be used to implement the GNN model when the required graph data and training setup are available.

The backend coordinates requests between the frontend, database, and recommendation component. The database stores available user and travel information, while external data sources can provide additional travel details when available.

The system can be developed incrementally, starting with the frontend and basic preference-based recommendations,

followed by backend integration, database connectivity, graph construction, and GNN-based ranking. The final description of the implementation and its results should reflect the components that have actually been developed and tested.

4. RESULTS AND DISCUSSION

The proposed TravelSense system presents a graph-based approach to personalized travel recommendation and adaptive itinerary planning. The research focuses on integrating user preferences, travel-related entities, and their relationships into a unified recommendation framework. Unlike conventional approaches that may recommend destinations independently, TravelSense aims to consider connections among destinations, hotels, activities, food places, events, and transportation options.

The proposed heterogeneous travel graph provides a structured representation of these relationships. By connecting users with travel entities through preferences and interactions, the system is designed to support more personalized recommendations. The integration of a Graph Neural Network (GNN) is intended to learn useful representations from the graph and rank candidate travel options according to their relevance to individual users.

Another expected outcome of the proposed system is the generation of travel itineraries based on user-specific constraints, including budget, travel duration, group size, and preferred activities. This approach aims to reduce the effort required to search for travel information across multiple sources and organize suitable options into a coherent travel plan.

The proposed framework also considers the cold-start problem. For users with limited interaction history, explicitly provided preferences can serve as initial inputs for generating recommendations. As additional interaction data becomes available, it may be incorporated into the recommendation process to improve personalization.

The effectiveness of the proposed approach must be established through implementation and experimental evaluation. Appropriate evaluation measures, such as Precision@K, Recall@K, and NDCG@K, can be used to assess recommendation quality when suitable test data is available. Itinerary feasibility can also be evaluated by checking compliance with budget and travel-duration constraints.

At the current stage, the described outcomes represent the intended capabilities and potential benefits of the proposed TravelSense framework rather than experimentally verified performance improvements. Quantitative results and comparisons with baseline recommendation methods will be reported after the corresponding experiments have been completed.

5. CONCLUSION

Using machine learning in tourism recommendation systems has a lot of benefits, but it also has a few small problems. But these problems can be solved by paying close attention to data quality, model design, and system security during the whole development process. Adding user feedback and looking into other ways to make recommendations can make them even more open and diverse. A well-designed machine-learning-based tourism recommendation system can help both travelers and travel companies in a big way. This makes it a promising technology with a lot of potential in the tourism industry. The future of machine learning-based tourism recommendation systems includes advanced recommendation algorithms for better suggestions, AI techniques for smart decision-making, real-time data sources for the most current information, personalized travel planning tools, integration with social media for user-generated content and collaboration, the creation of a mobile app for convenience, and the assurance of co These changes are meant to make the system more accurate, easier to use, and more useful for making suggestions for relaxing vacations and making users' overall travel experiences better. The potential for future advancements is to enhance the data collection process by integrating real-time data from wearable technology and smart environments. By doing so, it would be possible to provide personalized recommendations that are highly dynamic and can adapt to a traveler's current experiences and preferences. This would result in a continuous feedback loop that can offer an exceptionally customized travel experience.

6. REFERENCES

[1] Sarkar, Manali, Aparajita Das, Sraddha Roy Choudhury, and Siddhartha Chatterjee. "A* Based Optimized Travel Recommendation System for Smart Mobility." International Journal of Innovative Science and Research Technology 10, no. 5 (2025): 3185–3193.

[2] Afham, Mohammad, Vyom Pandey, and Golda Dilip. "Travel Recommendation System." International Journal of Advanced Research in Computer and Communication Engineering 14, no. 10 (2025): 466–470.

[3] Wang, Lang. "Travelling Route Recommendation Method Based on Graph Neural Network for Improving Travel Experience." Journal of Circuits, Systems, and Computers 32, no. 6 (2023): 2350102.

[4] Wicaksono, Adhika Putra, and Muhammad Izharuddin. "Understanding Gen Z's AI adoption for travel planning and recommendation intent." Manajemen dan Bisnis 25, no. 2 (2026): 405–423.

[5] Wang, Ching, and Yi-Shan Chu. "Intention Travel Recommendation System: A Travel Recommendation System based on Online Comments." In Proceedings of The IMA AI/ML Congress 2024, 2024.

[6] Christy, V. Maria, M. Mohamed Sameer Ali, S. Suman Rajest, and R. Regin. "Personalized Travel Recommendation System for Intelligent Tourism Planning." American Journal of Science on Integration and Human Development 4, no. 3 (2026): 1–17.

[7] Wu, Yi. "Analysis of the Impact of Tourists' Experiences, Review Ratings, and Budget Levels on Travel Recommendation Scores." In Proceedings of ICEMGD 2026 Symposium: Data-Driven Business Management, Corporate Governance, and Organizational Performance, 180–190, 2026.

[8] Aarif, K., J. Deepika, M. Ashwin Kumar, and P. Sanjana. "Deep neural collaborative filtering model for personalized travel recommendation." Scientific Reports 16, no. 1 (2026): 1–16.

[9] Turno, Francesco Maria, and Irina Yatskiv (Jackiva). "Graph-Based Approach for Personalized Travel Recommendations." Transport and Telecommunication 24, no. 4 (2023): 423–433.

[10] Karthiyayini, J., and R. J. Anandhi. "Personalized Travel Recommendations System Using Hybrid Filtering and Deep Learning." Preprint, Research Square (2024).